Introduction to Op-Amp

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Basic of Op-Amp

- A high-gain electronic voltage amplifier with a differential input and usually a single-ended output.
- A multistage amplifier in which a number of amplifier stages are interconnected to each other in a very complicated manner
- Its internal circuit consists of many transistors, FETs and resistors
- They are used extensively in signal conditioning, filtering or to perform mathematical operations such as add, subtract, integration and differentiation
- Widely used in a vast array of electronic devices.

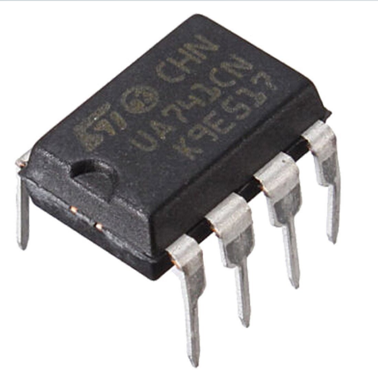


Fig: Op-Amp IC

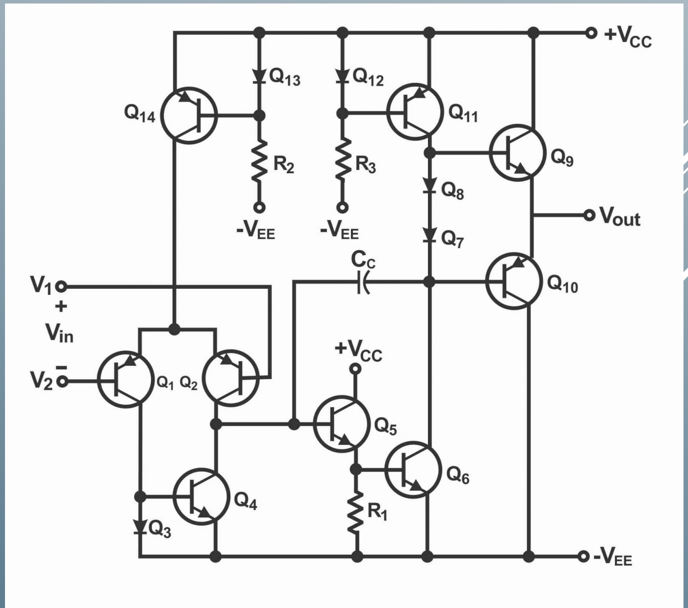


Fig: Op-Amp internal circuit

Op-Amp pin layout & description

- ✔ **Power Supply Pins: Pin 4 and 7**

  The negative and positive voltage power supply terminals. The voltage level between these pins can be in the range of 5 – 18V.

- ✔ **Output Pin: Pin 6**

  Voltage level at this pin indicates whether it's going to follow +ve supply voltage or –ve supply voltage.

- ✔ **Input Pins: Pin 2 and Pin 3**

  Pin 3 is considered as the inverting input while pin 3 is considered as the non-inverting input pin

- ✔ **Offset Null Pins: Pin 1 and Pin 5**

  An offset value of the voltage to be applied at pin 1 and pin 5, and this generally accomplished by a potentiometer.

- ✔ **Not Connected Pin: Pin 8**

  It has no connection with any of the internal or external circuits.

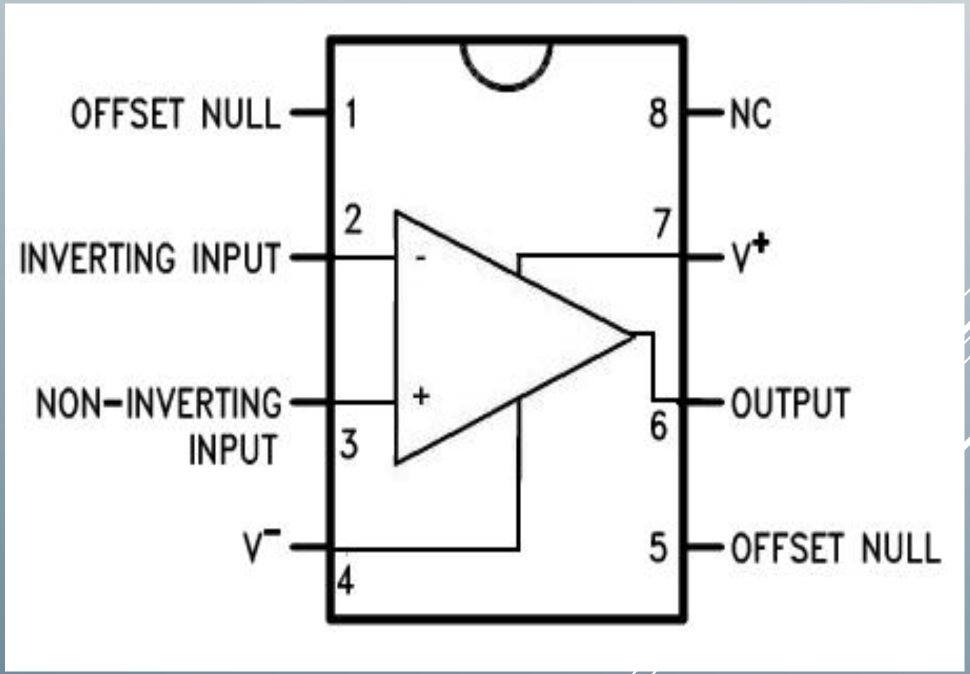


Fig: Op-Amp pin configuration

Op-Amp circuit symbol

- **Inverting Input (-), V_-**

The input terminal marked with a minus sign is the inverting input. If a signal is applied here, the output will be inverted (180-degree phase shift).

- **Non-Inverting Input (+), V_+**

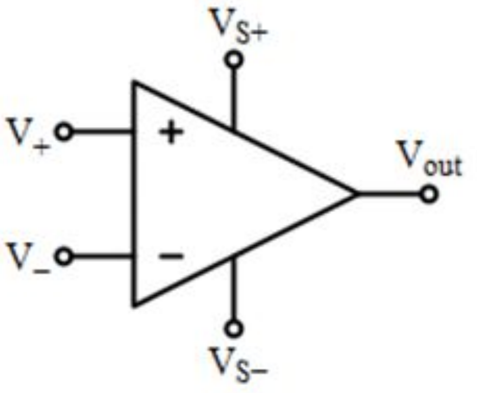
The input terminal marked with a plus sign is the non-inverting input. A signal applied here appears at the output with the same phase.

- **Output (V_{out})**

The pointed end of the triangle represents the output terminal, where the amplified signal is available.

- **Power Supply (V_{S+} and V_{S-})**

A standard op-amp symbol also includes terminals for the positive and negative power supplies, which are necessary for the amplifier to operate.



- V_+ : non-inverting input
- V_- : inverting input
- V_{out} : output
- V_{S+} : positive power supply
- V_{S-} : negative power supply

Fig: Circuit symbol of Op-Amp

Op-Amp characteristics features

Characteristic feature	Ideal Op-Amp	Practical Op-Amp
Open loop gain	Output voltage is infinitely large for even the smallest input voltage difference resulting in infinite gain.	Practical op-amps have very high open-loop gain (e.g., 200,000) but it is not infinite.
Differential input resistance	Infinite input resistance, so that almost any signal source can drive it and there is no loading of preceding stage.	Input resistance is typically 10^6 to 10^{12} ohm (for FET input Op-Amps such as uAF771), so still it draws some current and not all source can drive it.
Output resistance	Zero output resistance, so that output can drive an infinite number of other devices..	Output resistance is typically 75 ohm for standard Op-Amps, so it has limit to deliver current to output devices.
Bandwidth	The op-amp can amplify signals of all frequencies equally well.	Practical op-amps have a limited bandwidth, with the gain decreasing at higher frequencies.

Op-Amp characteristics features definitions

Open-loop gain

- The open-loop gain (usually referred to as A_{VOL}) is the gain of the amplifier without the feedback loop being closed, hence the name "open loop.
- In an ideal op amp, it is infinite with infinite bandwidth.
- While an ideal op-amp has infinite open-loop gain, real op-amps have very large, but finite, open-loop gains.
- Open-loop gain can be given by-

$$A = V_O / V_E ,$$

where $V_E = V_{IN+} - V_{IN-}$, is the difference between the voltage signals applied at its two input terminals.

- The open-loop gain affects the stability and accuracy of the op-amp circuit.

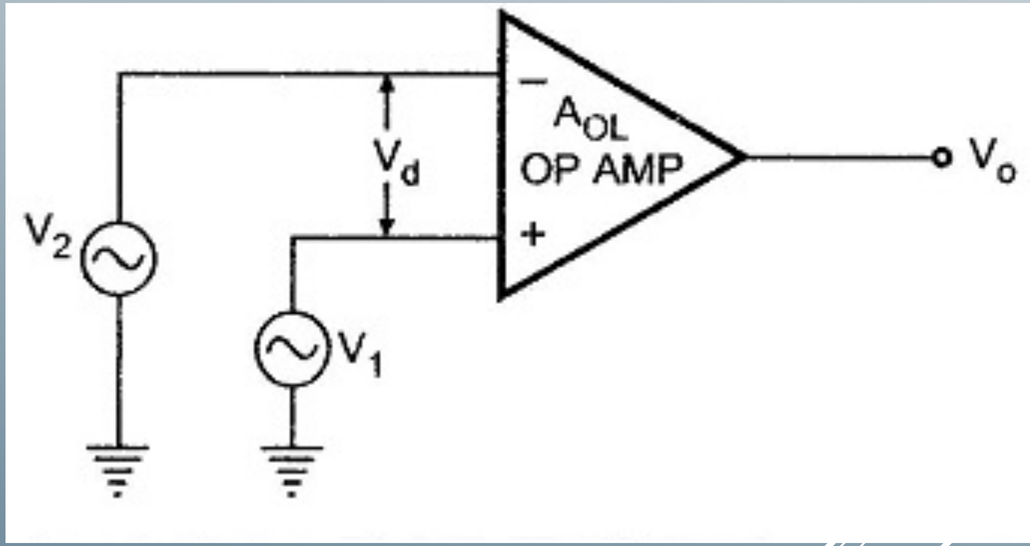


Fig: Open-loop configuration of Op-Amp

Op-Amp characteristics features definitions

Differential input resistance

- The differential input resistance of an operational amplifier (op-amp) is the resistance measured between its two input terminals.
- For an ideal op-amp, this resistance is considered to be infinite, meaning no current flows into the inputs.
- In practical op-amps, it's typically very high, often in the megaohms or even tera-ohms range.
- It minimizes the current drawn from the signal source connected to the op-amp's inputs. To reduce effect on the source's voltage and distorting the signal.
- The differential input resistance, $R_{diff} = R1+R3$

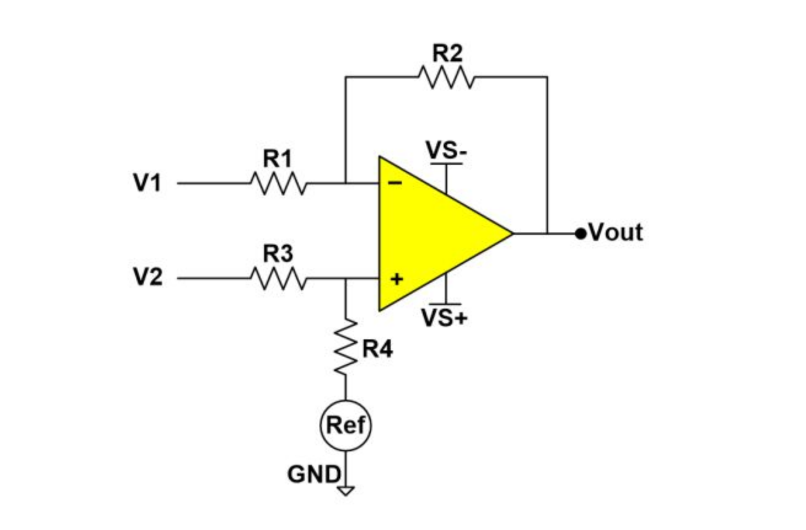
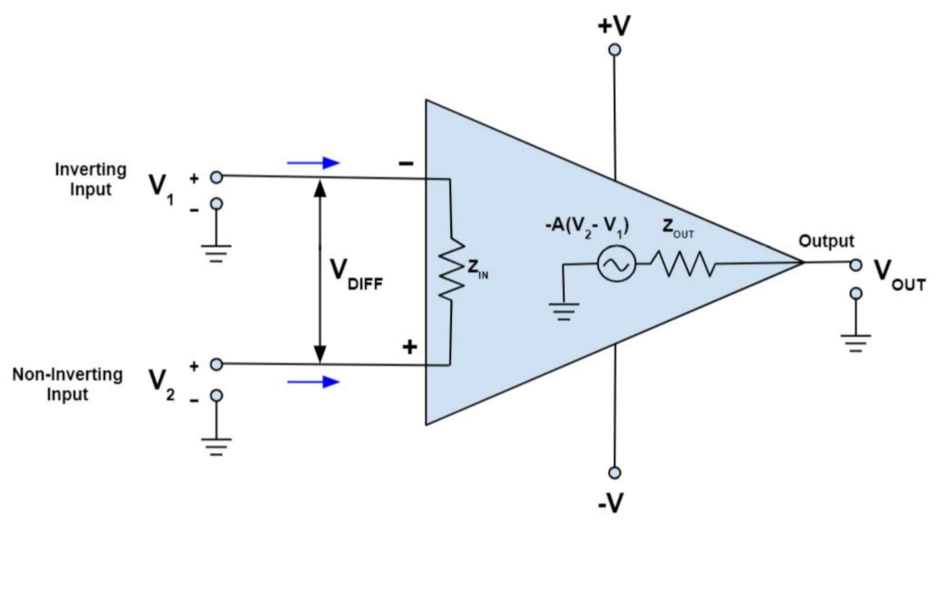


Fig: Differential input resistance calculation of Op-Amp

Op-Amp characteristics features definitions

Output resistance

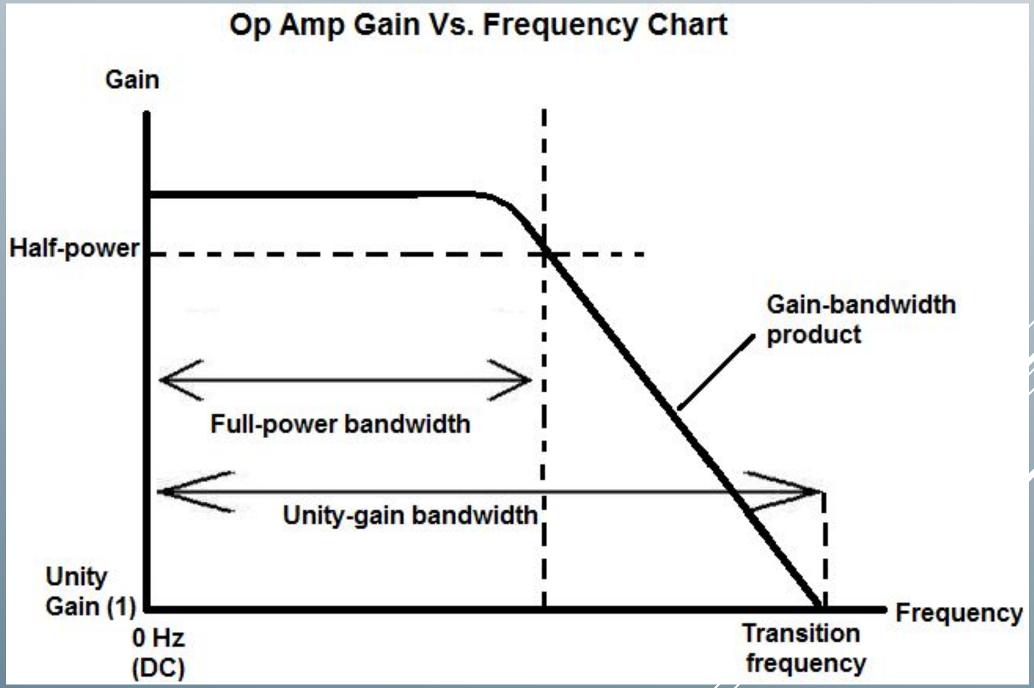
- The output resistance of the ideal operational amplifier is assumed to be zero acting as a perfect internal voltage source with no internal resistance so that it can supply as much current as necessary to the load.
- This internal resistance is effectively in series with the load thereby reducing the output voltage available to the load.
- Real op-amps have output impedances less than 100Ω without feedback and significantly lower with feedback.



Op-Amp characteristics features definitions

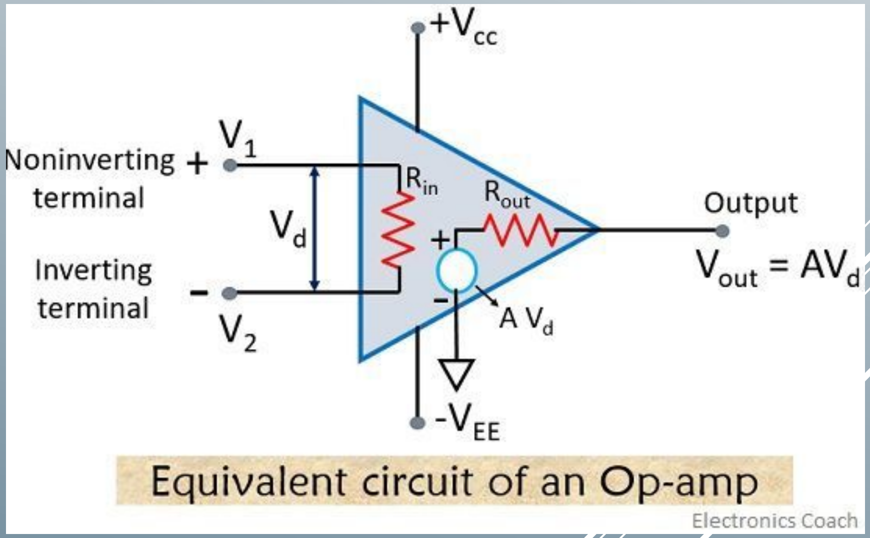
Bandwidth

- Bandwidth of an op-amp refers to the range of frequencies over which it can amplify a signal with a specified gain, typically measured at the -3 dB point
- An ideal operational amplifier has an infinite frequency response and can amplify any frequency signal from DC to the highest AC frequencies so it is therefore assumed to have an infinite bandwidth.
- With real op-amps, the bandwidth is limited by the Gain-Bandwidth product (GB), which is equal to the frequency where the amplifiers gain becomes unity.
- Op amps have limitations in their bandwidth due to internal capacitance and some other factors. Open-loop bandwidth is typically very limited, while closed-loop bandwidth can be much wider, especially at lower gain settings.



Op-Amp equivalent circuit

- **Differential Input Voltage (V_d),** the voltage difference between the non-inverting (+) and inverting (-) input terminals of the op-amp.
- **Input Resistance (R_{in}),** the internal resistance of the op-amp at its input terminals. In ideal op-amps, this is considered infinite, meaning no current flows into the input terminals.
- **Open-Loop Gain (A),** the amplification factor of the op-amp when no feedback is applied. It is the ratio of output voltage (V_{out}) to the differential input voltage (V_d): $A = V_{out} / V_d$.
- **Output Resistance (R_{out}),** the internal resistance of the op-amp at the output terminal.
- **Output Voltage (V_{out}),** the amplified voltage at the output of the op-amp, calculated as $V_{out} = A * V_d$.
- **Power Supply Terminals (V_{CC} and V_{EE}),** the positive and negative power supply voltages connected to the op-amp respectively.



Op-Amp looping configurations

Open-loop configuration

- No feedback, i.e., no connection exists between input and output terminals of any type
- Open-loop gain of the amplifier is much higher
- The output voltage is equal to the voltage gain, A times the difference between the two input voltages
- Due to the high gain and resulting instability, open-loop op-amps are generally not used for linear amplification of signals.
- Used in comparator and oscillator circuits
- Three open-loop configurations- i. Differential amplifier, ii. Inverting amplifier & iii. Non-inverting amplifier

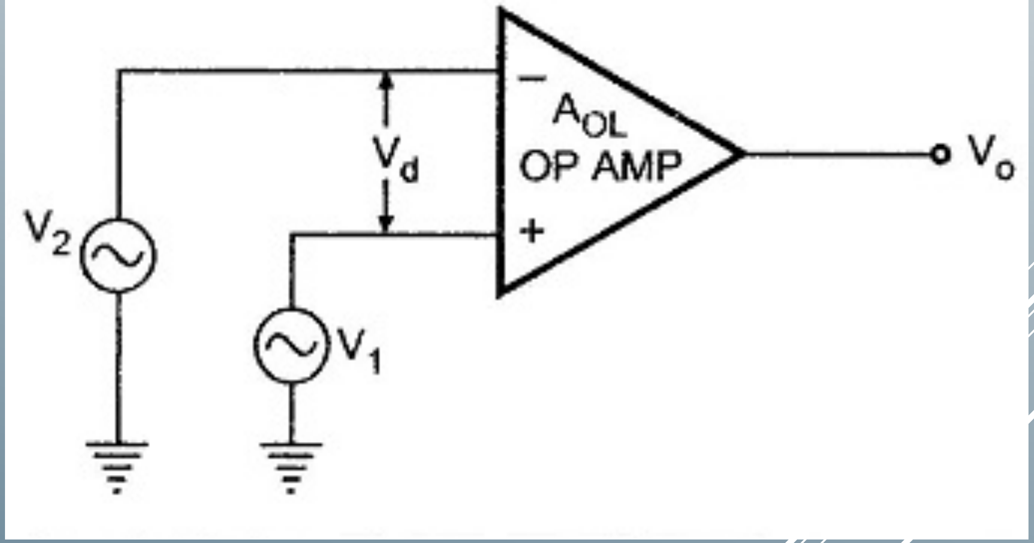


Fig: Open-loop configuration of Op-Amp

Op-Amp looping configurations

Close-loop configuration

- Applying feedback which means a portion of the output signal is returned to the input, to stabilize and control the amplifier's gain and other characteristics.
- This feedback is typically negative (degenerative), meaning the feedback signal opposes the input signal.
- Includes inverting and non-inverting amplifiers, each with distinct characteristics and applications.
- It allows precise control of the amplifier's gain using external resistors, making the gain less dependent on the op-amp's inherent open-loop gain.
- It can improve bandwidth, reduce distortion, and minimize the effects of temperature and power supply variations.
- The closed-loop gain is determined by the ratio of the feedback resistor (R_2) to the input resistor (R_1): Gain = $-R_2/R_1$.

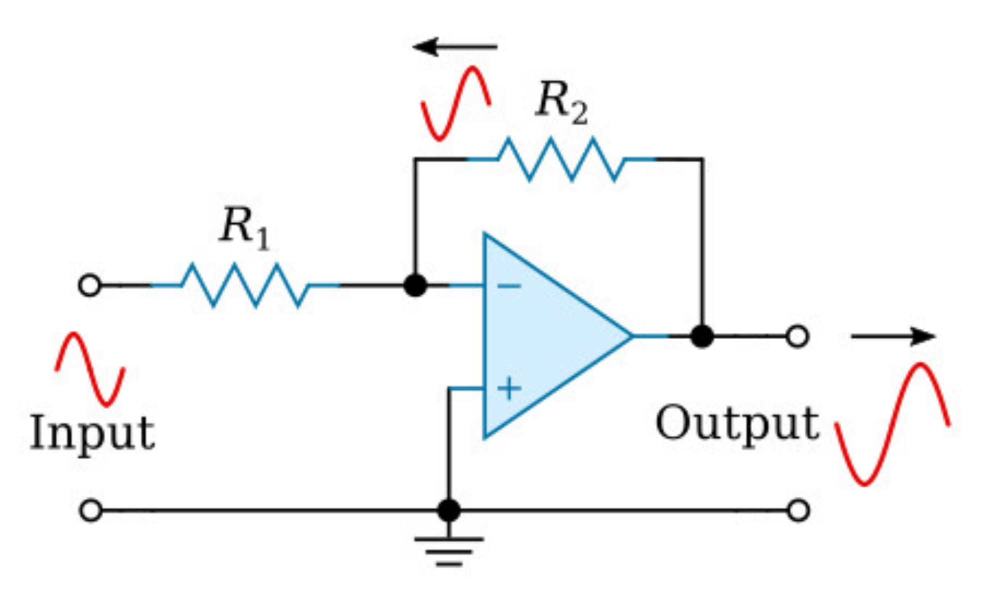


Fig: Close-loop configuration of Op-Amp

Op-Amp applications